

SIMULATION ANALYSIS OF BRAIN DRAIN PHENOMENA FROM INDONESIA USING SYSTEM DYNAMICS

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ABSTRACT

Recently, there have been Indonesian students who chose to study abroad with various initial intentions in increasing numbers. However, from the viewpoint of future national development it is feared that students finally may choose to stay abroad rather than to go back to Indonesia and to contribute to developing their homeland. Not only for students but also for professionals, especially specialists and engineers, feel more desirable to work in developed countries rather than their homeland. Although this phenomenon is not conspicuous, it is very significant to Indonesia. This study aims to grasp the general situation of 'Brain Drain' in Indonesia, to explain the 'Brain Drain' process by constructing a System Dynamics model, and to estimate the effects of government's policies by simulations under a variety of policy scenarios. Two main methodologies are used in this study. First is the use of descriptive statistics. The second methodology is application of System Dynamics (SD) to social model building. The questionnaire results suggest that 'education' factor is the most influential for respondent's decision on where to live; then followed by 'economy', 'social', 'environment', and 'politics' factors. The results also indicate that most of respondents' occupations are students. Second, according to the data gathered for calculating the correlations between factors, 'economic' indicator is considered as the core factor because it has high correlations with other factors except for 'social' indicator. Three mixed policy scenarios are selected for simulation and showing the most optimal policy scenario by comparison with the base scenario: (1) education and economy; (2) social and education; and (3) politics and economy. The result of simulations shows that each mixed policies has its own impact in different aspects because it has different characteristics and concentrations, hence it depends on the government which scenario to follow.

KEYWORDS

System Dynamics, students, professionals, government, mixed policies

1. INTRODUCTION

The increasing number of Indonesian students going abroad is not considered as a bad thing. However, in the long term future development the condition may change into unbeneficial phenomena to Indonesia, and it is caused by Indonesian students who choose to stay abroad than to go back to Indonesia. One of the reason that students may not go back home is because the

choice of major when they pursue their education. The lack of job vacancies in their fields of studies can be one of the major causes. The similar situation also happens for professionals, which can be the continuity of students remaining abroad, or self expatriate. Most Indonesian's talents work abroad specialize in certain occupations, usually scientists and researchers. One of the reasons why they choose to work abroad is because the lack of facilities of research provided by Indonesia government. It is supported by Hazen and Alberts' (2006) statement that some people move to advance countries to pursue higher education or conduct research in premier educational institutions and in sophisticated laboratories that do usually not exist in the sending country. Another reason is that the financial incentives for researchers, scientists, and lecturers are not as demanding as in other countries. Aside of that, Docquier (2006) analyzes that increasing number of skilled migrants is strongly related to increasing demographic sizes and severe rises in educational attainment in developing countries.

As many research have been done in the past, including Docquier's finding, the skilled emigration rate varies across countries, and depends on many factors such as: population size; political environment; policies in education; development level; and so on. So, many would ask "So how about in the future? Will they come back?" This is still a big question mark because so far there is no publicized research about brain drain in Indonesia. In other Asian countries, such as India and China, professionals having spent years of experience in United States and other developed countries decide to return back home. This is what is called brain circulation. In the end, although many talents are recorded going to work abroad, they will return home to contribute to their countries.

Therefore, this study aims to: To acknowledge the general condition of brain drain in Indonesia; to explain the brain drain process flow in Indonesia and clarify between students and professionals; to identify and estimate the amount of Indonesian students and professionals going abroad for 20 years; to identify estimated policies that can be implemented by Indonesian government to decrease the rate of brain drain or increase the rate of brain gain and brain circulation; and to determine the policies mixed that have the most optimal effect on decreasing the rate of brain drain or increasing the rate of brain gain and brain circulation.

2. LITERATURE REVIEW

Explored since 1960s, brain drain now is more far-reaching compared to around 25 years ago (Beine *et al.*, 2003). But some amount of mobility is obviously necessary if developing countries are incorporating into the global economy; and a large outflow of skilled people causes the threat of a brain drain, which can adversely impact local growth and development (Abella, 2001). According to Britain's Department for International Development (DfID) and International Labour Office's (ILO) study shows that several developing countries had lost up to 30% of their highly educated human capital (Lowell & Findlay, 2001). Adding developing countries' points of view, there seems to be several theories. First, the attractiveness for other countries to give foreign direct investments decreases if it is acknowledged that many talented people work abroad in particular developing countries. Specifically for scientists and researchers, they need same-level educated colleagues to make them feel comfortable and successfully obtain their objectives of their jobs. And when they cannot feel comfortable, likely they will try to find work abroad (Schelling, 1978). This also relates to the interdependence decisions that people make, which people try to consider previous cases of the same nationalities or others decisions and make it a collective action (Oliver, 1985). Although with this kind of principle, if there are more skilled workers stays in developing countries, then the rest of the elite members will follow.

Relating to structural factors, it appears that ethnic and professional factors dominate the factors for professionals to move abroad or stay; as for motivating factors, economic factors dominate the

‘push’ and ‘pull’ factors. Especially for scientists and specialists, the causes are mostly dominated by the need to develop themselves. Also they cannot keep aside low wages and salaries, worsening conditions (in general), the fact that the results of scientific activity have not been much in demand, low and further declining level of the prestige of science in society, the circumstance of vulnerability and lack of protection in the field of science, and the uncertain prospects for careers that influence scientists to go abroad, which makes sense since the spending on scientific research and experimental design work in a research conducted in Russia has added up only to 0.4-0.6 percent of the gross domestic product (Ushkalov and Malakha, 2001).

Considering the positive effects of migration for destination countries, there are two general effects on economic growth: wage effect; and talent and technology effect (Solimano, 2010). Also according to the recent research conducted by Docquier and Rappaport (2007) shows that “there is a positive correlation between emigrants to rich countries and the increase in the stock of human capital at home, albeit there is an exception whether those educated people will stay in home after graduation.” This study is one example that brain drain still has positive effects to people’s home countries.

3. METHODOLOGY

There are two main methodology used in this study: descriptive statistics and system dynamics. Descriptive statistics is used for several purposes. First, is to calculate the value score of factors (economy, social, environment, education, and politics). These factors are used just to identify which factor is the most influential or important for people to live in by calculating the means of every factor chosen by respondents about the importance of those factors. The second one is by analyzing the countries index to identify the high or low correlations between five factors.

System dynamics is a computer-aided approach to policy analysis and design. The conceptual tools and concept of the field (including feedback thinking, stock and flows, the concept of feedback loop dominance, and an endogenous point of view) are as important to the field as its simulation method. The use of system dynamics, in the end, is mainly to compare the base scenario where the policies have not been implemented, to the simulation when several mixed policies implemented into the model and which mixed policies come up with optimal result. Policy design includes the creation of entirely new strategies, structures, and decision rules.

4. RESULTS

4.1. Questionnaire Results

Respondents consist of 43% male, 52% female, and 5% unanswered. While from the group of age, respondents are dominated by 18-25 years old group (51%). To measure the importance of factors according to respondents, this research uses Likert scale, with 1 is the most important and 5 is the least important. To identify the score, mean (average) from each factor is measured, and it becomes the level of importance. Then the score is converted into the score 1 to be the least important and 5 to be the most important in order to avoid confusion.

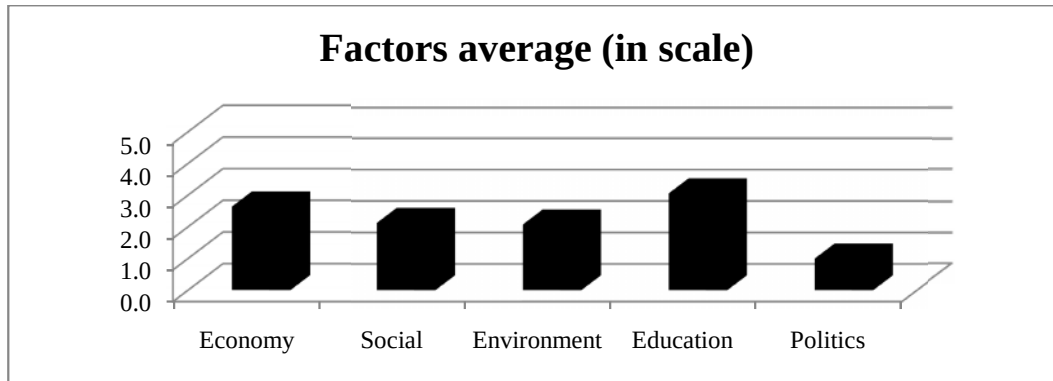


Figure 1. The importance of factors

The result shows that education is considered as the most important factor according to respondents. This result is attributed to the fact that respondents are mostly students. Nevertheless, some professionals put education condition first rather than economic condition. Hence, education factor or condition can be objectively measured as the most important. This result will not be inserted into the simulation, but in the recommendation to the government, this result can be suggested to represent from citizens' perspectives.

There are two valuable results from the open ended questionnaire that was asked to the respondents. One question is opportunities that respondents are looking for abroad that they cannot find in Indonesia, and second is advantages of studying abroad compared to in Indonesia. For the first question, they are several opinions worth to be noted. First, many of respondents answered that they were looking for experience, interaction, and competition with international people. For the first question, they are several opinions worth to be noted. First, many of respondents answered that they were looking for experience, interaction, and competition with international people. Second, for students taking specific majors that are not common in Indonesia, global networking can be a good opportunity for them to gain working experience, and it may happen if they study abroad. According to respondents, they are several advantages of studying abroad compared to in Indonesia. One example is the general quality of education; this includes the quality of education itself, the quality of lecturers, high research fund, high knowledge shares, and the facilities that universities have.

4.1. Countries Index

Five major factors considered in this research are economy, social, environment, education, and politics. The largest correlation is between politics and economy (0.785). This correlation may be attributed to the interdependence relationships between politics and economics condition in one country. For example, the stability of politic or security in a country affects the level of investments, the more stable political condition the higher foreign direct investment. A country that has many conflicts can discourage investors since the warranty of business success is questionable.

Table 1. Correlation matrix of countries index

		Social	Environment	Education	Politics	Economic
Correlation	Social	1.000	-.041	.583	.458	.327
	Environment	-.041	1.000	-.305	-.434	-.642
	Education	.583	-.305	1.000	.680	.603
	Politics	.458	-.434	.680	1.000	.785
	Economic	.327	-.642	.603	.785	1.000
Sig. (1-tailed)	Social		.364	.000	.000	.002
	Environment			.004	.000	.000
	Education				.000	.000
	Politics					.000
	Economic					

The implementation of correlations of factors will not be in the simulation, but as a set of example in the scenario analysis on the government policies. From this result, the most influencing factor is economic factor or condition. Therefore, it is best if the government considers how to apply effective economic policies and how they will affect other aspects.

4.3. Model Building

4.3.1. Population

In this model, the productive age is set from age 15 years old to 49 years old. The total population of productive ages then continues to universities enrollment.

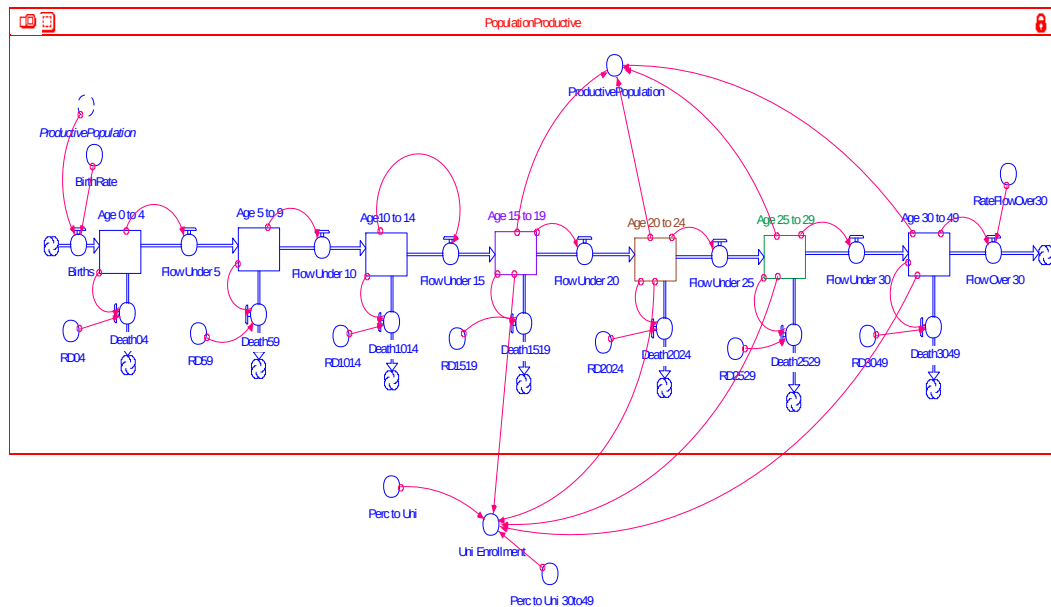


Figure 2. Population to universities enrolment

4.3.2. University enrolment

This stage explains about the university enrollment and how students make decisions whether to study in Indonesia or abroad. The basic flow is when students choose between domestic universities and international universities. The preliminary process before students going abroad is preparing for their departure. It includes preparation of language training, applying for students visa, university acceptance, financial guarantee, and so on. Therefore, not 100 percent of students who are willing to study abroad can actually go due to many requirements. Since this event usually happens, the drop outs from applying to study abroad need to be acknowledged in the model. If students drop out from studying abroad, then they will try to go apply for universities (flow of new students).

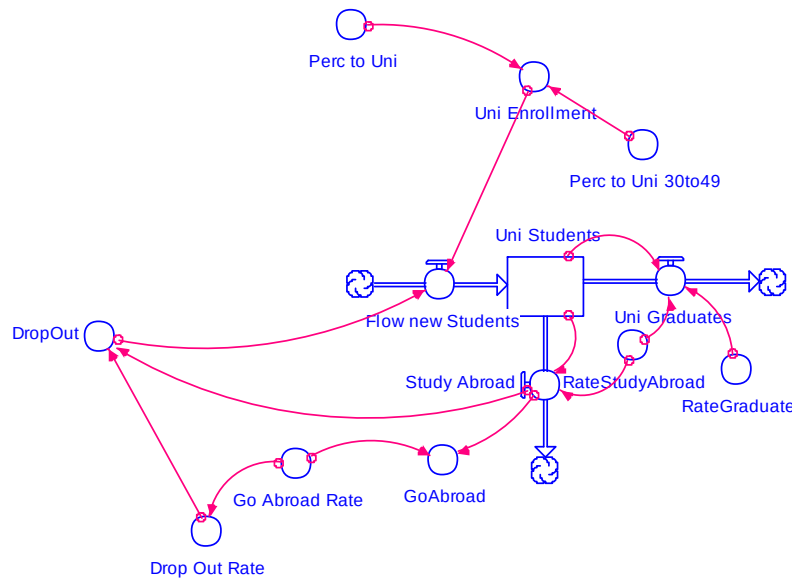


Figure 3. Flow of new university students and choice for their studies

4.3.3. Flow of students to selected countries

Students who manage to study abroad will spend their education time until they graduate. This section shows the flow of students going abroad to selected countries. The selected countries for the model are Australia, Canada, China, Germany, Japan, Netherlands, Singapore, United Kingdom (UK), and United States of America (USA).

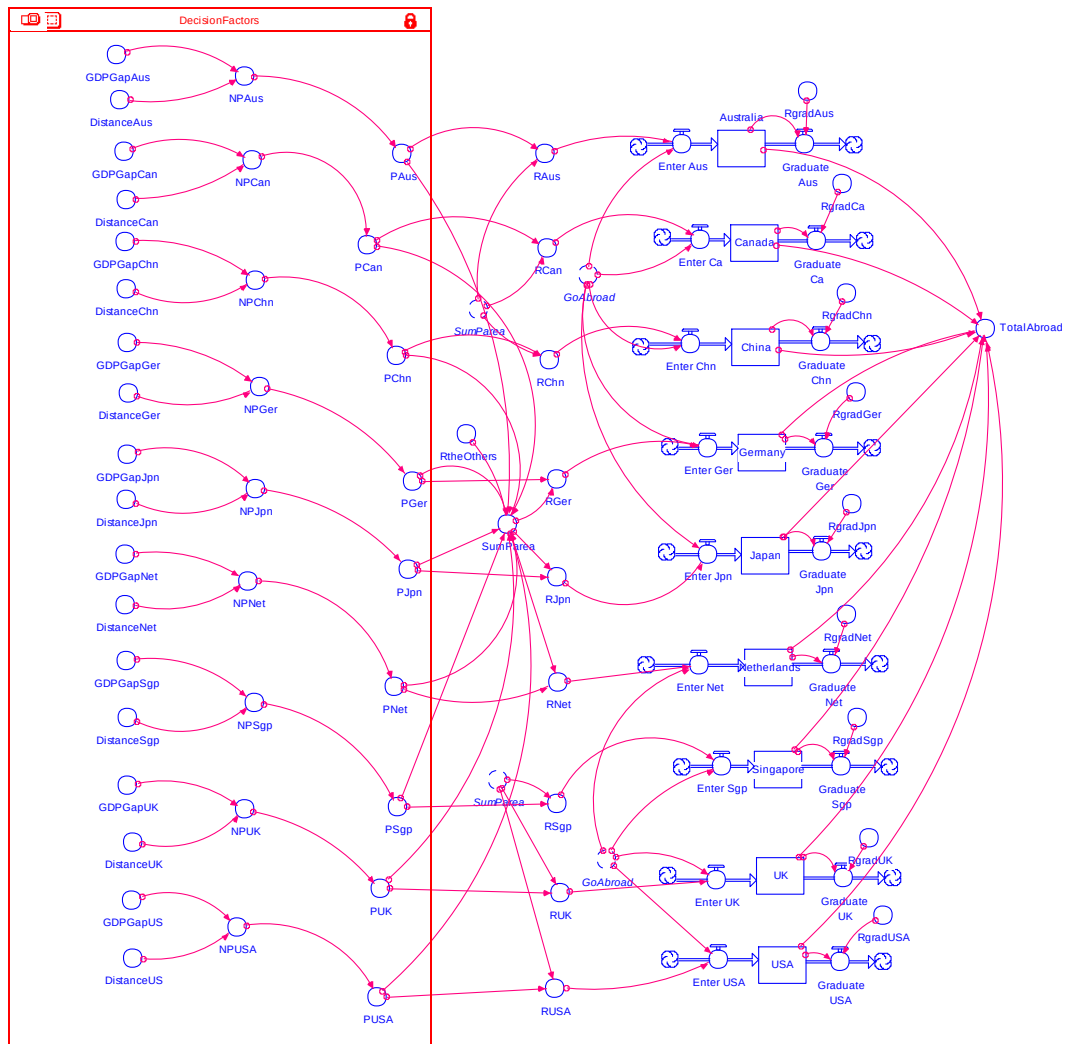


Figure 4. Flow of Indonesian students abroad

4.3.4. Flow after graduation

Graduates from abroad will be divided into three categories to specify the specialties: science and engineering (S&E); political and economics (P&Ec); and others. Based on major of study, this model can show the flow of graduates from each category from each country with their decision, whether to return home or remain abroad. This flow, later on, can be an inlet for government policies on which area of specialties that government feels to need the most.

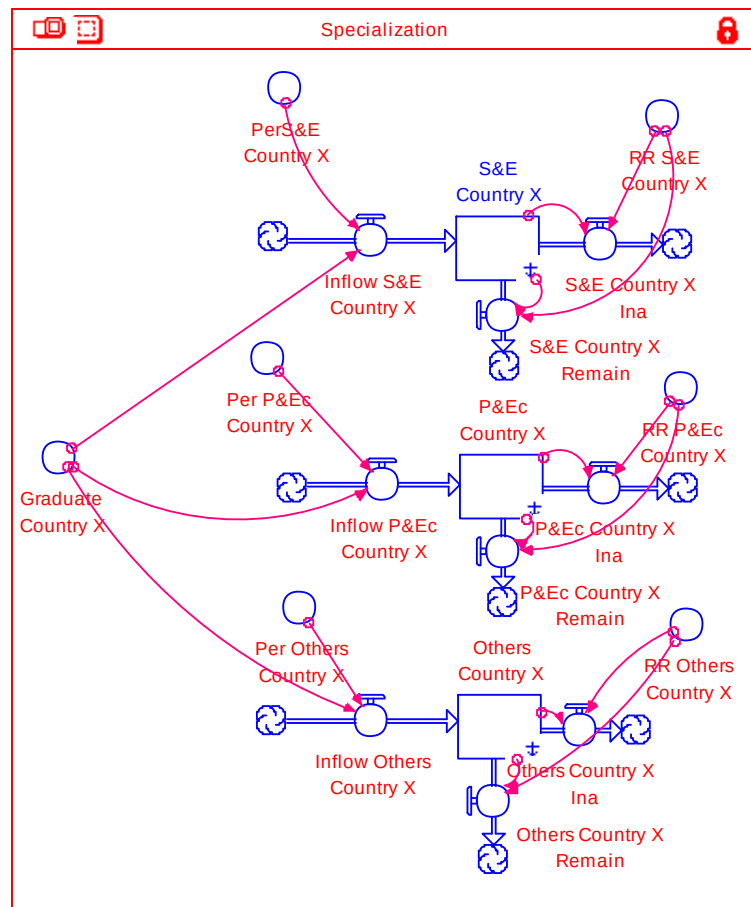


Figure 5. Flow specialization of students

4.3.4. Flow of graduates to professionals

From the previous flow, graduates are divided according to the major of their studies or their specialties. After that, three main majors of graduates who go back to Indonesia and who remain abroad will go to Talents in Indonesia (Talent Ina) and Talents abroad (Talent Outside). Meanwhile, the flow of professionals starts from Talent in Indonesia and Talents abroad to Professionals in Indonesia (Pro Ina) and Professionals abroad (Pro Abr). There are several paths of decisions for students after they graduate, as well as for existing professionals: (1) Students, who decide to return home, will go to talent pool of Indonesia, where they can choose whether to continue to work in Indonesia, or if they have chance to work abroad, they will choose to go abroad. It is assumed that students returning home are facing competitions with domestic universities graduates in order to find job not only in Indonesia, but also abroad; (2) Students, who decide to stay abroad, can keep staying there, or after some time they move back to Indonesia and become professionals in Indonesia; and (3) Professionals, whether in Indonesia and abroad, can move to other places (including Indonesia) or stay in countries they currently live.

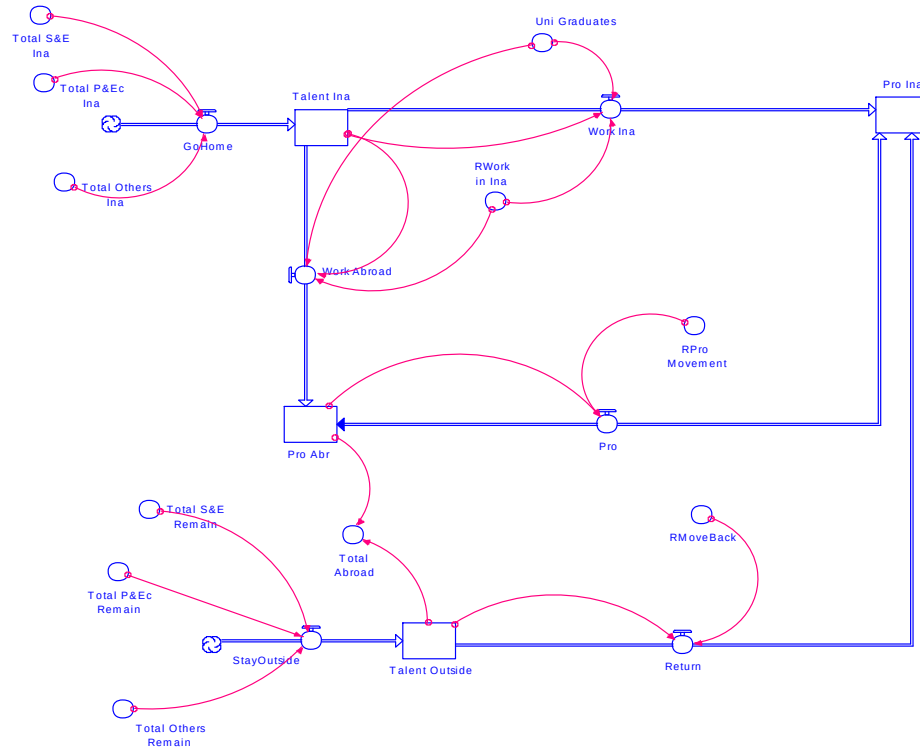


Figure 6. Flow of professionals in Indonesia and abroad

4.3.5. Base Simulation

The base model in this research is about the normal condition where there is no new policy implemented, only the existing policies planned at the beginning in 2012. The simulation will be conducted to set the benchmark for analyzing disparity of scenario settings that will be discussed later on this chapter. To make this possible, the first step is to set variables and assumptions. The variables required in this simulation are not easy to get, especially due to limited time. Nevertheless, with the general assumptions, the result of the simulation is useful as a reflection in Indonesia.

4.3.6. Students flow

First of all, the probability for productive population to enroll to universities is set previously with even number for each range of ages, except for 30-49 years old group since population in that age group has less percentage of enrollment compared to other age groups. In the simulation, flow of new students and the stock of university students increase gradually every year. This is due to the increase of population and also to the possibility of increasing economic condition of Indonesian citizens.

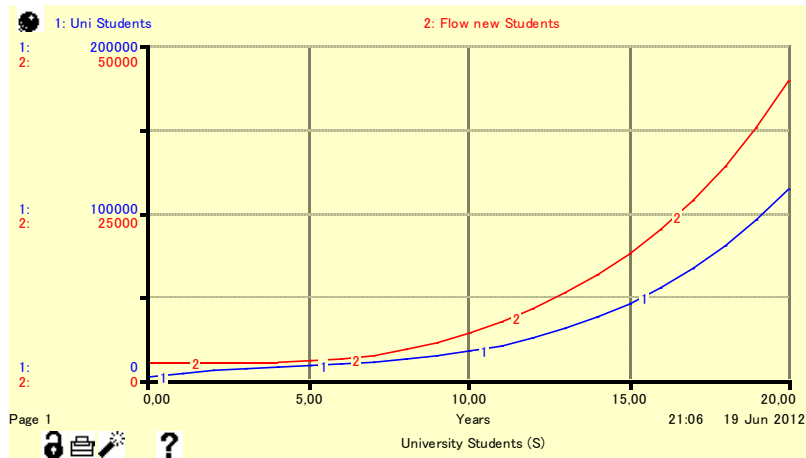


Figure 7. Estimation of university students and flow of new students

Figure 7 shows that within years ahead, amount of students who enroll to universities will get higher, hence increase the chance of students to go abroad. Flow of new students is little different from the increase of university students, since there are possibilities for students who decide to study abroad cannot fulfill the requirements and then have to repeat the process from the start.

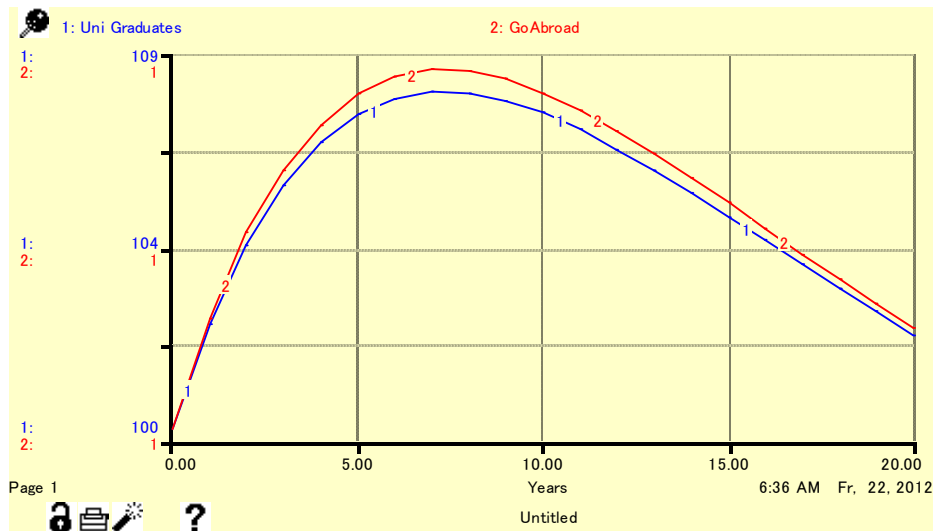


Figure 8. Comparison between total students abroad and students in Indonesia

Since the percentage of students who will study abroad has been set constantly from the beginning, the flow of students who go abroad and study in Indonesia do not have much of a difference. Figure 8 shows the level difference between students going abroad and students in Indonesia.

4.3.7. Professionals

In professionals' case, the important result in the end is the comparison between professionals abroad (Pro Abr) and professionals in Indonesia (Pro Ina). What needs to be paid attention to is

the scale of professionals abroad and professionals in Indonesia. The amount of professionals abroad reaches its peak drastically in the first year, and starts to decline after, and significantly.

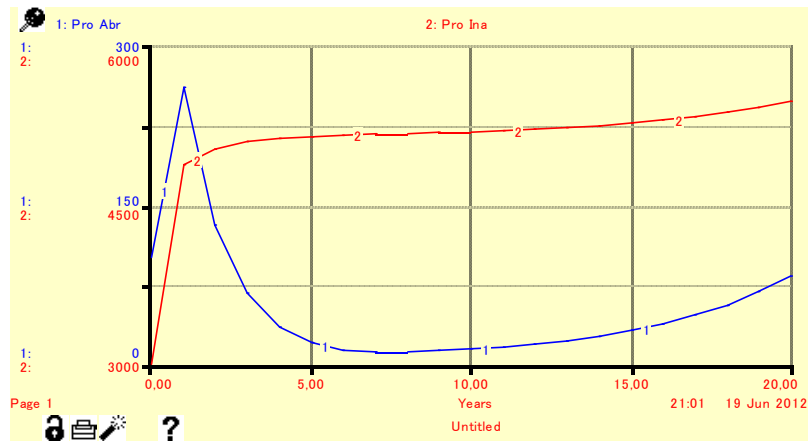


Figure 9. Professionals comparison (Indonesia and Abroad)



Figure 10. Comparison between talents in Indonesia, talents outside, professionals abroad, and professionals in Indonesia

Meanwhile, to compare trends of talents and professionals, figure 10 shows that students who decide to stay abroad increase, while students who decide to go home and work in Indonesia decrease rapidly. This can cause Indonesian human capitals decrease since many decide to work abroad than to go back home. However, need to be noted that looking at the scale in each graph, especially comparison between Indonesia and abroad, the amount of professionals abroad and/or talents outside Indonesia are considered small compared to professionals in Indonesia but if the simulated trend will occur in the future, the government needs to apply new policies in order not only to promote international universities graduates to return home, but also to increase the amount of professionals in Indonesia.

4.3.8. Policies scenario settings

In the base simulation, the trend results for Indonesian people to go abroad are worrisome. Therefore, in order to limit the flow, Indonesian government needs to implement effective and on target policies. However, the government cannot focus on this problem only, so there needs to be

cooperation from departments related to the policies that will be implemented by the government and mixed policies between two factors should be undergone in order to make it more efficient.

Table 3. Policies suggested

Sector	Policies suggested
Economic	Extra incentives for international graduates Increasing Foreign Direct Investment (FDI) Establishing "data bank"
Social	Promoting health sector Disease prevention programs Social campaign
Environment	Clean environment campaign Climate change anticipation program
Education	Increasing education quality in Indonesian universities Increasing research funds and salary for education sector workers Supporting technology innovations created by Indonesia's talented people
Politics	Increasing relationships with developed countries Developing a secure environment Improving and consistently increasing laws and regulations implementation

Therefore, this research set three scenarios of mixed policies: (1) Economic and education policies; (2) Social and education policies; (3) and Politics and economic policies. There are some assumptions made in the simulation. However, by making assumptions decision makers can set their own parameters that are close to the real condition and hence can decide which policies they should implement to produce the most optimal solution.

4.3.9. Policies scenario simulation analysis

The result of each scenario will be shown in a table below:

Table 4. Comparison between policy scenarios

Effects	Economic and education	Social and education	Politics and economic
Students	The amount of students going abroad increases, the return of international universities graduates is higher than graduates who remain abroad.	Slowly declining the amount of students who go abroad. The decreasing amount of students abroad also leads to decreasing amount of talents abroad.	The amount of students who go to Indonesian universities increases while students going abroad decreases.
Professionals	Professionals abroad the trend seems to be declining for the next 20 years.	Professionals abroad are noticed to be declining as professionals in Indonesia has inconsistent trend.	Professionals abroad also declines since people choose to work in Indonesia. Professionals in Indonesia are considered to have smaller impact by the policies.
Decrease flow of students abroad		√	√
Increase of talents going home	√		
Declining professionals abroad	√	√	√

5. CONCLUSION

There are several conclusions made based on this study. First, is that the brain drain phenomena in Indonesia is not new, and since the number is still small compared to other countries, the government does not pay too much attention to this matter. However, if the government does not do anything in particular to decrease the rate of brain drain, it is worrisome that Indonesia will lose many of its talented human capitals to developed countries in the future.

Second of all, based on the simulation of system dynamics, it can be concluded that of all three scenarios simulated, there is no best option of which the government should follow. This is mainly due to the uniqueness and the objectiveness from every scenario. And in the end, it depends on the government which mixed policy it wants to implement and which sectors that the government wants to focus on. Scenario one focuses on the economic and education sector in Indonesia, which results in higher students going abroad but at the same time higher students return because the stability of economic condition. With this condition, professionals abroad also tend to decrease. Scenario two focuses on social and education sectors, which in the end results in declining number of students going abroad and talents abroad. This is due to the government can increase the education quality in Indonesia. Scenario three focuses on politic and economic sectors, and the result of the simulation shows the decreasing students abroad and increasing students in Indonesia. Since the economic and political situation in Indonesia is strong, there are many investments implemented in Indonesia and professionals tend to choose to work in Indonesia instead of abroad.

6. RECOMMENDATION AND FURTHER RESEARCH

With three scenarios simulated, this research shows some alternatives for the government about problems related to brain drain. Nonetheless, this research cannot suggest which mixed policies that government should follow; because each mixed policies have its own uniqueness, objectives,

and concentrations. And since conducting this research has limited time and data, this research is still in premature stage, and it needs more concrete data in order to forecast the condition and trends that will be close to the reality. However, by using this model, one can follow the model and then add necessary additional factors or variables, since the condition in every country is different.

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